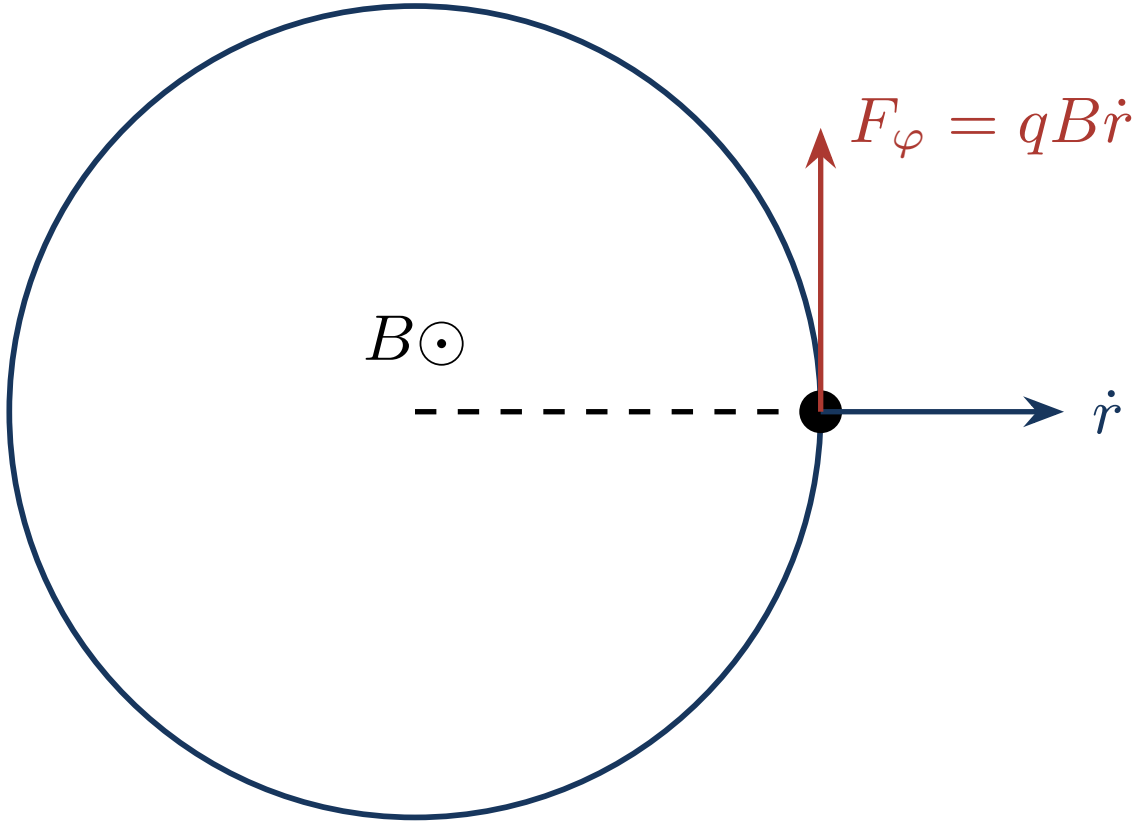

外场向纸外，径向速度产生切向磁力；$\dot{L}_z = qBr\dot{r}$